

Documentation

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1 Steps

In this document you will find the sequential steps that made themis walk, these steps will be tied together on a mathematical basis.

- Isaac Sim directly resets articulation to its configured initial state. After one physics step, actuator ans damping are set to zero, from then on the QP sends joint effort commands.
- The code reads joint angles, speed, position, and body speed. (It uses this to info to calculate the COM and foot position, FK for the foot position and for the com :

$$c = \frac{\sum_i m_i p_i}{\sum_i m_i}$$

- Via the virtual encoders, contact sensors measure how much vertical force acts at the heel, inner toe, and outer toe. It approximates them :

$$F_{Rz} = \left| F_{Rz}^{\text{heel}} + F_{Rz}^{\text{inner}} + F_{Rz}^{\text{outer}} \right|, \quad F_{Lz} = \left| F_{Lz}^{\text{heel}} + F_{Lz}^{\text{inner}} + F_{Lz}^{\text{outer}} \right|.$$

- The previous information is useful for the initial pose, targets are created from the initial pose. The planner selects the stance foot: Left and swing foot right.
- It stores the initial ankle midpoint;

$$p_{R0,x} = \text{initial } x\text{-position of the right foot}$$

$$p_{L0,x} = \text{initial } x\text{-position of the left foot}$$

Therefore,

$$x_{\text{mid},0} = \frac{p_{R0,x} + p_{L0,x}}{2}$$

represents the initial midpoint between the right and left feet along the x -axis. From this point the planner finds the position of the next foot. The next right foot is then computed RF [1],

$$\mathbf{RF}[k+1] = \begin{bmatrix} x_{\text{mid},0} + (k+1)(0.10) \\ y_{\text{lane}} \\ z_{\text{ground}} - 0.004 \end{bmatrix} \quad (1)$$

Where K is initially zero, notice that we subtract 0.004 m from the ground height to trick the robot, this allows a stronger grip to the ground.

- every (1/60) seconds, the planner produces desired COM, DCM, ankle, load, and contact values.

- the QP converts COM errors to desired accelerations:

$$\ddot{\mathbf{p}}_s^* = \ddot{\mathbf{p}}_s^d + \mathbf{K}_p (\mathbf{p}_s^d - \mathbf{p}_s) + \mathbf{K}_d (\dot{\mathbf{p}}_s^d - \dot{\mathbf{p}}_s), \quad (2)$$

$$\ddot{\mathbf{c}}^* = \ddot{\mathbf{c}}^d + \mathbf{K}_{p,c} (\mathbf{c}^d - \mathbf{c}) + \mathbf{K}_{d,c} (\dot{\mathbf{c}}^d - \dot{\mathbf{c}}) + \begin{bmatrix} 8(\xi_x^d - \xi_x) \\ 8(\xi_y^d - \xi_y) \\ 0 \end{bmatrix} \quad (3)$$

$$\mathbf{K}_{p,c} = \text{diag}(40, 40, 80), \quad \mathbf{K}_{d,c} = \text{diag}(12, 12, 18),$$

$$\mathbf{K}_{p,s} = \text{diag}(80, 80, 120), \quad \mathbf{K}_{d,s} = \text{diag}(18, 18, 22).$$

$$\mathbf{e}_B = \log [\mathbf{R}_B^d \mathbf{R}_B^T]^\vee, \quad \dot{\boldsymbol{\omega}}_B^* = \mathbf{K}_{p,B} \mathbf{e}_B - \mathbf{K}_{d,B} \boldsymbol{\omega}_B \quad (4)$$

$$\mathbf{e}_{R,s} = \log [\mathbf{R}_s^d \mathbf{R}_s^T]^\vee, \quad \dot{\boldsymbol{\omega}}_s^* = \mathbf{K}_{p,s} \mathbf{e}_{R,s} - \mathbf{K}_{d,s} \boldsymbol{\omega}_s \quad (5)$$

$$\mathbf{K}_{p,B} = \text{diag}(60, 60, 20), \quad \mathbf{K}_{d,B} = \text{diag}(14, 14, 8),$$

$$\mathbf{K}_{p,s} = \text{diag}(60, 60, 35), \quad \mathbf{K}_{d,s} = \text{diag}(15, 15, 10).$$

- These errors are the main parameters, that carry the significant weight in the objective function:

$$\begin{aligned} \mathcal{J}(\dot{\boldsymbol{\nu}}, \boldsymbol{\tau}, \mathbf{f}) = & 1000 \left\| \mathbf{J}_c \dot{\boldsymbol{\nu}} + \dot{\mathbf{J}}_c \boldsymbol{\nu} - \ddot{\mathbf{c}}^* \right\|^2 \\ & + 500 \left\| \mathbf{E}_B \dot{\boldsymbol{\nu}} - \dot{\boldsymbol{\omega}}_B^* \right\|^2 \\ & + 1000 \gamma_s \left\| \mathbf{J}_{s,v} \dot{\boldsymbol{\nu}} + \dot{\mathbf{J}}_{s,v} \boldsymbol{\nu} - \ddot{\mathbf{p}}_s^* \right\|^2 \\ & + 1000 \gamma_s \left\| \mathbf{J}_{s,\omega} \dot{\boldsymbol{\nu}} + \dot{\mathbf{J}}_{s,\omega} \boldsymbol{\nu} - \dot{\boldsymbol{\omega}}_s^* \right\|^2 \\ & + 50 \left\| \mathbf{A}_{G,\text{ang}} \dot{\boldsymbol{\nu}} + \left(\dot{\mathbf{A}}_G \boldsymbol{\nu} \right)_{\text{ang}} + 2 \mathbf{h}_{G,\text{ang}} \right\|^2 \\ & + 5 \left\| \mathbf{E}_q \dot{\boldsymbol{\nu}} - [5 (\mathbf{q}^{\text{nom}} - \mathbf{q}) - 3 \dot{\mathbf{q}}] \right\|^2 \\ & + 2 \times 10^{-3} \left\| \mathbf{f} - \mathbf{f}^{\text{ref}} \right\|^2 + 10^{-4} \left\| \dot{\boldsymbol{\nu}} \right\|^2 + 10^{-6} \left\| \boldsymbol{\tau} \right\|^2 \\ & + \mathbf{1}_{\{\boldsymbol{\tau}^- \text{ exists}\}} 10^{-5} \left\| \boldsymbol{\tau} - \boldsymbol{\tau}^- \right\|^2 \\ & + 10^{-7} \left\| \mathbf{f} \right\|^2 + 2 \times 10^{-5} \left\| \mathbf{f} - \mathbf{f}^- \right\|^2 \\ & + 10^{-10} \left(\left\| \dot{\boldsymbol{\nu}} \right\|^2 + \left\| \boldsymbol{\tau} \right\|^2 + \left\| \mathbf{f} \right\|^2 \right). \end{aligned} \quad (6)$$

- Having these weights:

Table 1: Runtime weights used in the objective function.

Term	Runtime weight
COM/DCM acceleration	1000
Base orientation	500
Swing position	$1000\gamma_s$
Swing orientation	$1000\gamma_s$
Centroidal angular momentum	50
Joint posture	5
Force-reference tracking	2×10^{-3}
Acceleration regularization	10^{-4}
Torque regularization	10^{-6}
Torque-change regularization	10^{-5}
Force regularization	10^{-7}
Force-change regularization	2×10^{-5}
Numerical regularization	10^{-10}

- The constraints, are distributed by priority too:

$$\mathbf{H}(\mathbf{q})\dot{\boldsymbol{\nu}} + \mathbf{CG}(\mathbf{q}, \boldsymbol{\nu}) = \mathbf{S}^T \boldsymbol{\tau} + \mathbf{J}_p(\mathbf{q})^T \mathbf{f} \quad (7)$$

$$|f_{x,i}| \leq 0.60 f_{z,i}, \quad |f_{y,i}| \leq 0.60 f_{z,i}. \quad (8)$$

For each point on an active foot,

$$0 \leq f_{z,i} \leq 250 \text{ N}. \quad (9)$$

For every point on an inactive foot,

$$f_{x,i} = f_{y,i} = f_{z,i} = 0. \quad (10)$$

Torque magnitude and change

$$-\bar{\boldsymbol{\tau}} \leq \boldsymbol{\tau} \leq \bar{\boldsymbol{\tau}}. \quad (11)$$

For each leg,

$$\bar{\boldsymbol{\tau}}_{\text{leg}} = \begin{bmatrix} 60 & 60 & 120 & 120 & 80 & 60 \end{bmatrix}^T \text{ N m}. \quad (12)$$

For each arm joint and the head joint,

$$\bar{\tau}_{\text{arm}} = 40 \text{ N m}, \quad \bar{\tau}_{\text{head}} = 10 \text{ N m}. \quad (13)$$

After the first control tick,

$$\boldsymbol{\tau}^- - 15 \text{ N m} \leq \boldsymbol{\tau} \leq \boldsymbol{\tau}^- + 15 \text{ N m}. \quad (14)$$

When a foot becomes active, its measured pose $(\mathbf{p}_A, \mathbf{R}_A)$ is captured. With componentwise clipped errors,

$$\mathbf{e}_p = \text{clip}(\mathbf{p} - \mathbf{p}_A, -0.02, 0.02), \quad \mathbf{e}_R = \text{clip}\left(\left[\text{Log}\left(\mathbf{R}_A \mathbf{R}^T\right)\right]^\vee, -0.15, 0.15\right). \quad (15)$$

$$\mathbf{a}_A^* = \begin{bmatrix} -100\mathbf{e}_p - 20\mathbf{v}_A \\ 80\mathbf{e}_R - 18\boldsymbol{\omega}_A \end{bmatrix}. \quad (16)$$

$$\mathbf{J}_A \dot{\boldsymbol{\nu}} + \dot{\mathbf{J}}_A \boldsymbol{\nu} = \mathbf{a}_A^*. \quad (17)$$

There are six equality rows per active foot: three translational and three rotational.

- Eventually the QP solves for the acceleration and forces, which in return will solve for the torques and check whether they are valid or not via the equality and the inequality constraints, and then these torques will be sent to the robot which will actuate the joints. Hence the solution vector of the optimization problem has this form:

$$\mathbf{X} = \begin{bmatrix} \ddot{\mathbf{C}} \\ \boldsymbol{\tau} \\ \mathbf{f} \end{bmatrix} \in R^{34+28+18} = R^{80}. \quad (18)$$

In total we have 28 joints which makes it 28 torques, but accounting for the floating base dynamics we need 6 more DOF hence 34 accelerations, and regarding the forces on each sole we have the heel, inner, and outer toe. Each of these point has a 3-D force accounting for FX, FY, and FZ. So for each foot we have a total of 9 forces, and for the both that makes it 18.

2 Planner

As the planner is considered a significant part of this methodology, a section is dedicated to explain the detailed process that goes throughout the planner.

2.1 Exact Measured Planner Inputs

At control step k , the planner receives

$$\mathbf{y}_k = \{t_k, \mathbf{c}_k, \dot{\mathbf{c}}_k, \mathbf{R}_{B,k}, \boldsymbol{\omega}_{B,k}, \mathbf{p}_{R,k}, \dot{\mathbf{p}}_{R,k}, \mathbf{R}_{R,k}, \mathbf{p}_{L,k}, \dot{\mathbf{p}}_{L,k}, \mathbf{R}_{L,k}, F_{Rz,k}, F_{Lz,k}\}. \quad (19)$$

These are:

- t_k : current planner time.
- $\mathbf{c}_k, \dot{\mathbf{c}}_k$: measured COM position and velocity.
- $\mathbf{R}_{B,k}, \boldsymbol{\omega}_{B,k}$: base orientation and angular velocity.
- $\mathbf{p}_{R,k}, \dot{\mathbf{p}}_{R,k}, \mathbf{R}_{R,k}$: right ankle position, velocity, and orientation.
- $\mathbf{p}_{L,k}, \dot{\mathbf{p}}_{L,k}, \mathbf{R}_{L,k}$: left ankle position, velocity, and orientation.
- $F_{Rz,k}, F_{Lz,k}$: total measured vertical force beneath each foot.

2.2 PREPARE-DS

Both feet remain on the ground while the robot stabilizes. The planner changes to **SHIFT** after at least 1.0 s when the following conditions remain continuously satisfied for 0.15 s:

$$F_R + F_L \geq 0.75W = 278.63 \text{ N}, \quad (20)$$

$$F_R \geq 0.10W, \quad F_L \geq 0.10W, \quad (21)$$

$$\|\dot{\mathbf{c}}_{xy}\| \leq 0.15 \text{ m/s}, \quad (22)$$

$$|\text{roll}| \leq 7^\circ, \quad |\text{pitch}| \leq 7^\circ. \quad (23)$$

Thus, the planner first confirms that both feet support the robot and that the body is stable.

2.3 SHIFT

The desired COM moves toward the stance foot. Both feet remain active contacts.

The planner changes to **LIFT** after at least 1.5 s when

$$\frac{F_{\text{stance}}}{F_R + F_L} \geq 0.80, \quad (24)$$

$$\frac{F_{\text{swing}}}{F_R + F_L} \leq 0.20, \quad (25)$$

$$F_{\text{swing}} < 0.05W = 18.58 \text{ N}, \quad (26)$$

$$\|\mathbf{c} - \mathbf{c}_{\text{shift}}^d\| \leq 0.04 \text{ m}, \quad (27)$$

together with

$$\|\dot{\mathbf{c}}_{xy}\| \leq 0.15 \text{ m/s}, \quad \text{tilt} \leq 7^\circ. \quad (28)$$

The measured DCM must also be inside the stance-foot support region.

These conditions must remain true for 0.15 s.

2.4 LIFT

The swing ankle moves upward while only the stance foot remains an active contact.

The planner changes to **TRANSFER** after at least 0.8 s when

$$F_{\text{swing}} < 18.58 \text{ N}, \quad (29)$$

and the measured ankle has risen by at least

$$p_{\text{swing},z} - p_{\text{start},z} \geq 0.020 \text{ m}. \quad (30)$$

These conditions must remain true for 0.08 s.

2.5 TRANSFER

The swing ankle moves horizontally toward the next foothold while remaining at its raised height.

The planner changes to **LOWER** after at least 1.2 s when

$$\|\mathbf{p}_{\text{swing},xy} - \mathbf{p}_{\text{target},xy}\| \leq 0.015 \text{ m}, \quad (31)$$

and

$$|p_{\text{swing},z} - p_{\text{apex},z}| \leq 0.015 \text{ m}. \quad (32)$$

Therefore, the foot is not lowered until it is horizontally close to its target.

2.6 LOWER

The swing ankle descends toward the ground.

The planner changes to **TOUCHDOWN** when the sensor detects

$$F_{\text{swing}} > 0.001W = 0.372 \text{ N}, \quad (33)$$

and the measured landing pose satisfies

$$\|\mathbf{p}_{\text{swing},xy} - \mathbf{p}_{\text{target},xy}\| \leq 0.020 \text{ m}, \quad (34)$$

$$|p_{\text{swing},z} - p_{\text{target},z}| \leq 0.015 \text{ m}, \quad (35)$$

$$e_R \leq 3^\circ. \quad (36)$$

There is no additional touchdown debounce in the current runtime configuration.

2.7 TOUCHDOWN

Both feet become active contacts again, and the load is gradually transferred onto the newly landed foot.

After at least 1.5 s, the step is accepted when

$$F_{\text{new}} \geq 0.20W = 74.30 \text{ N}, \quad (37)$$

$$F_{\text{old}} \geq 0.20W = 74.30 \text{ N}, \quad (38)$$

$$F_{\text{new}} + F_{\text{old}} \geq 0.75W = 278.63 \text{ N}. \quad (39)$$

The landing pose must also remain within the same position and orientation tolerances.

These conditions must remain true for 0.15 s. The planner then performs

$$\text{step index} \leftarrow \text{step index} + 1, \quad (40)$$

$$\text{new stance foot} \leftarrow \text{landed foot}, \quad (41)$$

$$\text{new swing foot} \leftarrow \text{opposite foot}, \quad (42)$$

and returns to PREPARE_DS for the following step.

Therefore, the full gait loop is

<pre> PREPARE_DS → SHIFT → LIFT → TRANSFER → LOWER → TOUCHDOWN → PREPARE_DS. </pre>	(43)
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